```
import pandas as pd
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
dataframe=pd.read_csv('/content/drive/MyDrive/Colab Notebooks/Data Science/Dataset/housing.csv')
```

```
dataframe.head() # print first 5 rows
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_valu
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200

```
dataframe.tail()
```

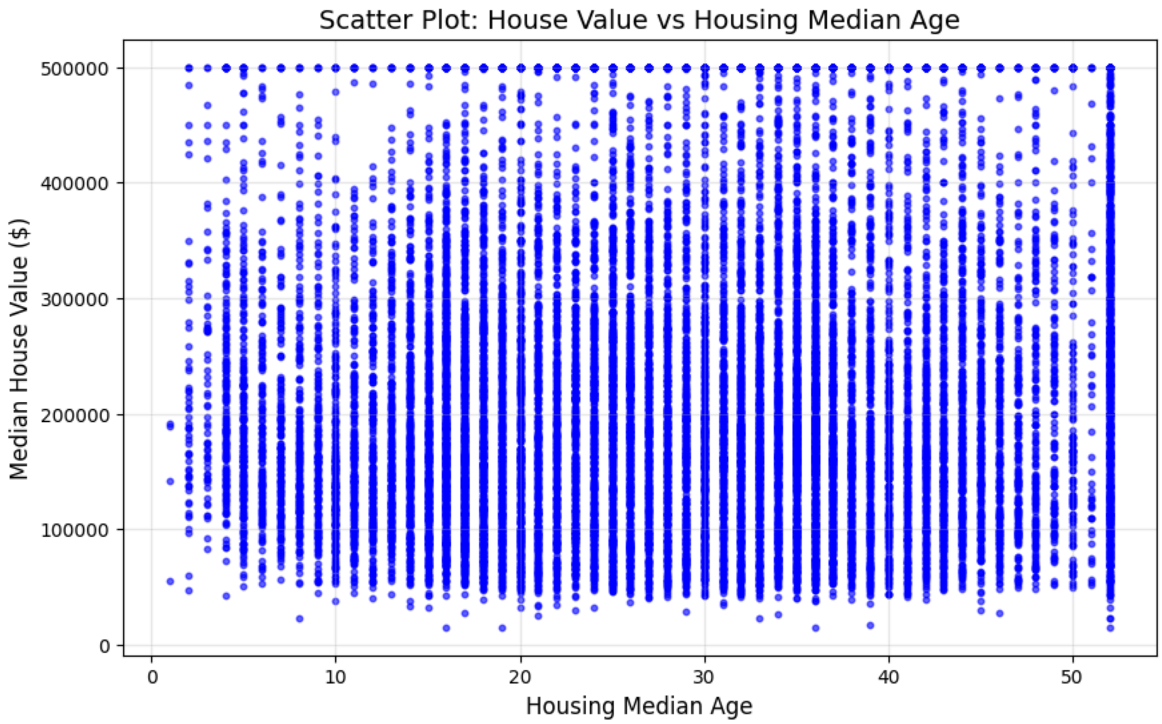
	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_
20635	-121.09	39.48	25.0	1665.0	374.0	845.0	330.0	1.5603	7
20636	-121.21	39.49	18.0	697.0	150.0	356.0	114.0	2.5568	7
20637	-121.22	39.43	17.0	2254.0	485.0	1007.0	433.0	1.7000	9
20638	-121.32	39.43	18.0	1860.0	409.0	741.0	349.0	1.8672	8
20639	-121.24	39.37	16.0	2785.0	616.0	1387.0	530.0	2.3886	8

```
dataframe.columns
```

```
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value', 'ocean_proximity'],
      dtype='object')
```

```
import matplotlib.pyplot as plt
import pandas as pd

# Create scatter plot
plt.figure(figsize=(10, 6))
plt.scatter(dataframe['housing_median_age'], dataframe['median_house_value'],
            alpha=0.6, color='blue', s=10)
plt.xlabel('Housing Median Age', fontsize=12)
plt.ylabel('Median House Value ($)', fontsize=12)
plt.title('Scatter Plot: House Value vs Housing Median Age', fontsize=14)
plt.grid(True, alpha=0.3)
plt.show()
```




```
print(dataframe.dtypes)
```

```
longitude             float64
latitude              float64
housing_median_age    float64
total_rooms           float64
total_bedrooms        float64
population            float64
households            float64
median_income         float64
median_house_value    float64
ocean_proximity        object
dtype: object
```

```
matrix = dataframe.to_numpy()

print("Matrix shape:", matrix.shape)  # (n_samples, n_features)
print("Matrix type:", type(matrix))   # numpy.ndarray
print(matrix[:3])  # First 3 rows as numpy array
```

```
Matrix shape: (20640, 10)
Matrix type: <class 'numpy.ndarray'>
[[-122.23 37.88 41.0 880.0 129.0 322.0 126.0 8.3252 452600.0 'NEAR BAY']
 [-122.22 37.86 21.0 7099.0 1106.0 2401.0 1138.0 8.3014 358500.0
  'NEAR BAY']
 [-122.24 37.85 52.0 1467.0 190.0 496.0 177.0 7.2574 352100.0 'NEAR BAY']]
```

```
dataframe.to_numpy()
```

```
array([[-122.23, 37.88, 41.0, ..., 8.3252, 452600.0, 'NEAR BAY'],
       [-122.22, 37.86, 21.0, ..., 8.3014, 358500.0, 'NEAR BAY'],
       [-122.24, 37.85, 52.0, ..., 7.2574, 352100.0, 'NEAR BAY'],
       ...,
       [-121.22, 39.43, 17.0, ..., 1.7, 92300.0, 'INLAND'],
       [-121.32, 39.43, 18.0, ..., 1.8672, 84700.0, 'INLAND'],
       [-121.24, 39.37, 16.0, ..., 2.3886, 89400.0, 'INLAND']],
      dtype=object)
```

```
# Feature matrix (independent variables)
X = dataframe[['longitude', 'latitude', 'housing_median_age', 'total_rooms',
              'total_bedrooms', 'population', 'households', 'median_income']].to_numpy()

# Target vector (dependent variable)
y = dataframe['median_house_value'].to_numpy()

print("X shape:", X.shape)  # (n_samples, 8 features)
```